

The Name of the Title Is Hope

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Abstract

LLM agents with web access are increasingly becoming the default for user-facing question answering. Reliable evaluation in this setting requires tests that remain unseen at train time and cannot be answered via verbatim lookup at inference. However, standard practice still relies on static benchmarks that age quickly and are vulnerable to contamination. We analyze two channels, pretraining leakage where test items appear in the pretraining data, and run-time leakage where the agent finds the exact target answer verbatim during evaluation and reports it rather than demonstrating genuine problem solving. We uncover evidence across five QA datasets for both pretraining and run-time leakage, and we show that static evaluation can overestimate capability. We introduce DynamicWebQA, a dynamic evaluation framework that constructs questions and verifiable answers at evaluation time from relevant web sources. The framework enforces multi-document grounding, records evidence chains, and offers controllable complexity through source set size, number of hops, and reasoning constraints. By generating fresh test instances, DynamicWebQA reduces memorization risk, mitigates run-time leakage, and naturally covers time-sensitive queries.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper;** *Generate the Correct Terms for Your Paper;* Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 Introduction

Benchmark leakage refers to the contamination that occurs when a model has prior exposure to a benchmark’s test items during pretraining, leading to inflated evaluation scores and undermining the benchmark’s validity [? ?]. Since most LLMs are pretrained indiscriminately on publicly available web content, there is a high likelihood that they might have encountered benchmark datasets hosted on platforms like HuggingFace, GitHub, Kaggle, and similar

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sites. If the base LLM used in a WebQA agent has already seen the test examples during pretraining, the evaluation effectively becomes a case of testing on the training set — severely compromising the credibility of the reported results.

Dynamic benchmarking is an emerging evaluation paradigm that generates or adapts test samples at evaluation time, so models are scored on fresh, unseen data and testset contamination is avoided [? ? ? ? ?]. That means the agent does not see the same test sample twice, effectively addressing issues such as LLM memorization.

In this work, we refer to information that can change over time, such as the stock price of Tesla, as evolving information. Arguably, this type of content could be labeled as dynamic content or temporal content, but that might cause confusion, since we are already using the term “dynamic” to describe an evaluation paradigm (dynamic benchmarking), and “temporal” might lead readers to associate it with time-series data. To avoid such confusion, we adopt the term evolving information to specifically denote factual content that may shift over time without implying a particular format or evaluation framework.

2 Related Work

Open-Domain QA (ODQA). Open-domain question answering studies how to find and use evidence from very large corpora to answer free-form questions. A sequence of datasets marked key shifts in task design: SQuAD [?] established span-based reading comprehension; Natural Questions [?] moved to real search queries; TriviaQA [?] stressed compositional, trivia-style questions with substantial lexical and syntactic variation; and HotpotQA [?] formalized multi-hop reasoning across multiple documents with sentence-level supporting facts and comparison questions, while including distractors to test *focus under noise*. Models such as fusion-in-decoder (FiD) [?] showed that aggregating many retrieved passages is a powerful way to synthesize scattered evidence.

Another important dimension in QA is *time*. RealTime QA [?] and FreshQA [?] evaluate whether systems can answer what is true *right now*, revealing failure modes when indices are stale or retrieved content is outdated or contradictory. These settings highlight needs that go beyond span extraction: recognizing when the answer has changed, when a question carries a false premise, and when recent evidence is required over older but higher-ranked documents.

Finally, surface simplicity does not imply task simplicity. SimpleQA [?] demonstrates that short, seemingly straightforward questions can still require precise world knowledge or multi-step reasoning, indicating that difficulty is not well represented by length or phrasing.

Retrieval-Augmented Generation (RAG). Modern RAG approaches follow a common pipeline – given a query, retrieve top- k passages from a non-parametric memory and condition a generator on them. Foundational formulations include RAG [?], REALM [?], RETRO [?], and ATLAS [?], which together demonstrate that query-time retrieval improves knowledge-intensive tasks while enabling the external memory to be updated without re-training the parametric model. Recent surveys study components and variants of this paradigm [?].

In the literature, RAG is distinguished from broader tool-augmented generation – RAG requires *query-conditioned retrieval* whose contents directly condition the generator. Calculators, code execution, or static always-prepended context are typically treated as non-retrieval tools [?]. In practice, the retrieval index can be the open web or enterprise/local corpora, and the mechanism can be lexical (e.g., BM25) or dense; the generator then aggregates evidence across multiple passages [?].

Beyond single-shot RAG, agentic frameworks let models iteratively issue searches, follow links, and cite sources. WebGPT augments an LLM with browsing and requires collecting references during answering [?]. ReAct further couples reasoning traces with tool calls (e.g., search, calculator) in a unified prompting loop, enabling multi-hop planning and verification [?]. These approaches often embed RAG steps inside a broader plan-act-observe loop, allowing complex queries that require cross-document synthesis.

Robustness of QA Systems. High IID (in-distribution) scores do not guarantee reliability under distribution shift [?]. Adversarial studies show that small, misleading perturbations can derail

readers, revealing reliance on shallow cues [?]. BEIR further documents pronounced *cross-domain* drops for dense retrievers and reader pipelines and highlights BM25's robustness, with no single model performing uniformly well across 18 heterogeneous datasets [?]. In realistic open-domain settings, models must also handle noisy or conflicting evidence across retrieved documents. Recent analyses quantify the prevalence of conflicts on the web (~25% of factoid questions) and show that current LLM-based QA systems often choose incorrectly or average contradictions [?]. Complementary retrieval-aware stress tests such as RARE introduce controlled perturbations at both the query and document levels and reveal brittleness under document changes and multi-hop settings [?].

Temporal robustness is equally critical: world knowledge changes faster than model weights. Time-sensitive and streaming benchmarks stress whether systems can keep up with new information, including WikiData/Wikipedia-derived temporal QA [?], news-stream evaluation in StreamingQA [?], the continually refreshed RealTime QA [?], and FreshLLMs/FreshQA [?]. Across these settings, retrieval augmentation substantially helps but also introduces new failure modes (e.g., latency, stale indices, contradictory sources).

Data Contamination. Data contamination occurs when evaluation items—or close paraphrases and answer-bearing spans—appear in the corpora used for pretraining, instruction tuning, or fine-tuning. Under contamination, test behaviour can collapse into lookup of memorized strings rather than the intended inference or problem solving, inflating leaderboard scores and masking brittleness [?]. Empirical work connects such leakage to memorization dynamics: larger models memorize more and earlier in training; duplication and longer prompts amplify verbatim regurgitation; and model capacity allows substantial unintended storage of datapoint-specific information [???]. At the same time, verbatim completion is not conclusive evidence of membership—models can sometimes perfectly complete sequences that were not explicitly seen during training—so naïve n -gram or completion-based membership tests can be misleading [?]. Moreover, the metric impact of contamination is dataset- and model-dependent [?].

Ravaut et al. survey detection methods and distinguish settings where candidate corpora are available (*open-data*) versus unavailable (*closed-data*) [?]. Open-data methods include instance- and dataset-level *string matching* (substrings, n -grams, edit distance), *embedding similarity*, and *paraphrase detection*. Closed-data methods include dataset-level *performance analysis* (train-test gaps, controlled retraining, temporal comparisons), *membership-inference attacks* (reference-based and reference-free), and instance-level probes of *model memorization* (e.g., prompt engineering, cloze tasks) and *model confidence* (likelihood or hidden-layer probing).

Despite limitations, string/span matching remains the workhorse in practice. It is simple, auditable, and the only approach that routinely scales to multi-billion-document crawls; when training data are inaccessible for closed models, behavioural signals alone cannot certify contamination [?]. Li et al. further note that most prior analyses rely on substring matching and that, without training-set access, uncovering contamination from model behaviour is extremely difficult [?].

For retrieval-augmented and agentic QA, pretraining-only audits are insufficient. At evaluation time, systems may retrieve and quote target items from web or enterprise indices, producing *search-time data contamination* that boosts scores without demonstrating reasoning [?]. Even low apparent rates can reorder competitive leaderboards and erode trust when online access is allowed [?]. These observations motivate evaluation protocols that pair scalable contamination checks with designs that limit both pretraining exposure and evaluation-time leakage [? ? ?].

Dynamic Benchmarking. Most benchmarks for agentic or retrieval-augmented QA remain static and publicly accessible (e.g., HotpotQA), which makes them susceptible to contamination and memorization [? ? ? ?]. As models and training corpora grow, leakage can inflate reported capability and mask brittleness. This has motivated *dynamic* evaluation paradigms that resist memorization and better reflect real-world, evolving information needs. Early platforms like Dynabench and Dynaboard supported human-and-model-in-the-loop data collection and hosted live evaluations [? ? ?]. Other works instantiate dynamic settings in specific environments (e.g., AndroidWorld for interactive agents) [?] and in time-sensitive QA streams.

Graph-based and procedural generation offer additional control over complexity and evidence diversity. For structured knowledge, Dynamic-KGQA generates fresh yet distributionally consistent QA sets from knowledge graphs, enabling reproducible but non-static test variants [?]. However, KG updates are periodic, and web-grounded dynamic QA remains under-explored. Our work targets this gap by constructing evaluation instances at test time from current web sources while recording evidence chains and enforcing multi-document grounding.

3 Method

We propose a method that utilizes *seed graphs* to anchor future question generation instead of relying on static datapoints that are susceptible to memorization or data contamination. Because each evaluation round renders a fresh QA pair from each graph, no fixed test item can be leaked ahead of time, yet all rounds remain semantically comparable via the common anchors.

3.1 Seed Graphs

At evaluation round t , we maintain a set $\mathcal{S}_t = \{G_1^{(t)}, \dots, G_{n_t}^{(t)}\}$ of seed graphs. Each graph encodes the evidence trail (e.g., queries, retrieved documents, answer node) for a distinct information need. To support evolving information, graphs are constructed at evaluation time.

3.2 QA Generation

The generation function Gen takes a seed graph $G_i^{(t)}$, a random seed $r_{i,t} \sim \mathcal{R}$, and a fixed generation configuration to produce one QA pair:

$$(q_{i,t}, a_{i,t}) = \text{Gen}(G_i^{(t)}; r_{i,t}, \theta) \quad (1)$$

Here, Gen is an LLM-based generator that bundles a frozen prompt template and fixed decoding settings (e.g., temperature, top- p , stop tokens). $r_{i,t}$ is an i.i.d. sample from randomness source $\mathcal{R}$, introducing per-sample stochasticity. Holding θ constant and

varying only $r_{i,t}$ ensures that each system sees independent but identically distributed QA samples.

3.3 Distributional Consistency

Sampling an independent seed $r_{i,t} \sim \mathcal{R}$ for each graph yields the per-graph distribution:

$$\mathcal{P}_{i,t} = \text{Dist}(\text{Gen}(G_i^{(t)}; r, \theta)) \quad (2)$$

Because seeds are independent across graphs, the round- t dataset $D_t = \{(q_{1,t}, a_{1,t}), \dots, (q_{n_t,t}, a_{n_t,t})\}$ is an i.i.d. sample from the product measure:

$$\mathcal{P}_t = \prod_{i=1}^{n_t} \mathcal{P}_{i,t} \quad (3)$$

For a model M under evaluation, let $\text{score}((q, a), M)$ be any per-question metric (e.g., exact match, F1, log-loss). The aggregate dataset score is defined as:

$$\text{Agg}(D_t, M) = \frac{1}{n_t} \sum_{i=1}^{n_t} \text{score}((q_{i,t}, a_{i,t}), M) \quad (4)$$

By linearity of expectation and independence of $(q_{i,t}, a_{i,t}) \sim \mathcal{P}_{i,t}$:

$$\mathbb{E}_{D_t \sim \mathcal{P}_t} [\text{Agg}(D_t, M)] = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{E}_{(q,a) \sim \mathcal{P}_{i,t}} [\text{score}((q, a), M)] \quad (5)$$

This shows that the expected accuracy is simply the average of the per-graph expectations—so every seed graph contributes equally.

3.4 Collision Bound

Assume for every seed graph $G_i^{(t)}$, the generator can produce K distinct question-answer pairs:

$$C_{i,t} = \{(q, a) \in \mathcal{P}_{i,t} \mid \Pr((q, a)) > 0\}, \quad |C_{i,t}| = K$$

To characterize overlap across rounds $u < v$, define:

$$J_{i,u,v} = |C_{i,u} \cap C_{i,v}|, \quad J_{\max} = \max_{u < v} J_{i,u,v}$$

Because answers and evidence evolve over time, $J_{\max}$ tends to remain small for fast-changing topics.

Collision Probability. Sampling one QA pair per round, the probability that seed graph i produces the same QA pair twice within t rounds satisfies:

$$\Pr[\text{collision within } t] \leq \frac{t(t-1)}{2K^2} J_{\max} \quad (6)$$

Collision Control. To guarantee that this probability is at most $\delta \in (0, 1)$, it suffices to choose K such that:

$$K \geq \sqrt{\frac{t(t-1)}{2\delta}} J_{\max} \quad (7)$$

Equation (7) captures key trade-offs:

- Larger K (richer candidate pool) reduces collision probability for fixed t .
- Fewer evaluation rounds t allow a smaller K for the same error tolerance δ .
- Smaller $J_{\max}$ (i.e., greater semantic drift) reduces required K .

Because each seed graph is rebuilt from fresh evidence each round, its candidate set changes, helping keep $J_{\max}$ small. Equation (6) quantifies the collision risk, while Equation (7) provides a tunable design guideline for selecting K .

3.5 Cross-Round Reporting and Compatibility

Because each round rebuilds the seed graph set S_t , the underlying distribution $\mathcal{P}_t = \prod_i \mathcal{P}_{i,t}$ may drift over time, especially for evolving topics. Consequently, raw accuracies from different rounds may not be directly comparable. We offer two strategies:

i) *Snapshot Evaluation*. Fix a reference round t^* and freeze its seed graphs S_{t^*} . All systems are then evaluated on the same dataset D_{t^*} , ensuring identical test conditions.

ii) *Macro-Averaged Score*. For longitudinal tracking, aggregate performance across a window $\{t_1, \dots, t_T\}$ (e.g., most recent T rounds) using:

$$\text{Score}_{\text{macro}} = \frac{1}{T} \sum_{j=1}^T \text{Agg}(D_{t_j}, M) \quad (8)$$

Each D_{t_j} is i.i.d. from its own $\mathcal{P}_{t_j}$, so Eq. (8) summarizes average effectiveness under the benchmark's natural evolution, without letting any single round dominate.

4 Results

5 Discussion

TBD

6 Conclusion

TBD

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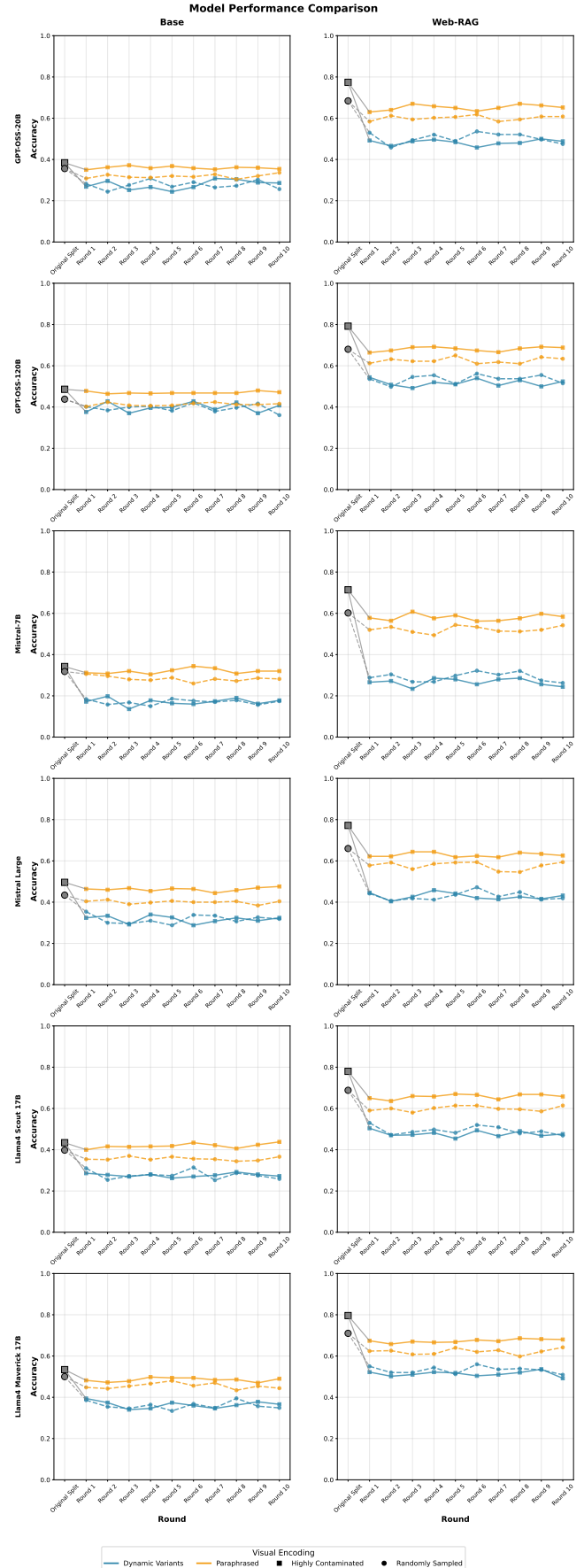


Figure 1: model_performance_collage.pdf

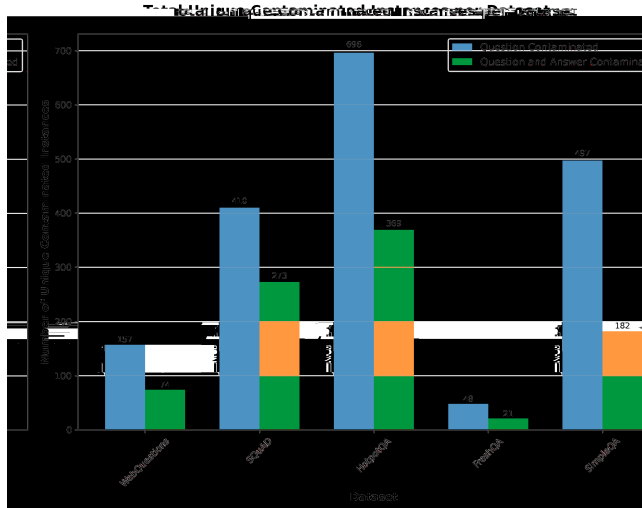


Figure 2: total_unique_contaminated_instances_per_dataset.png

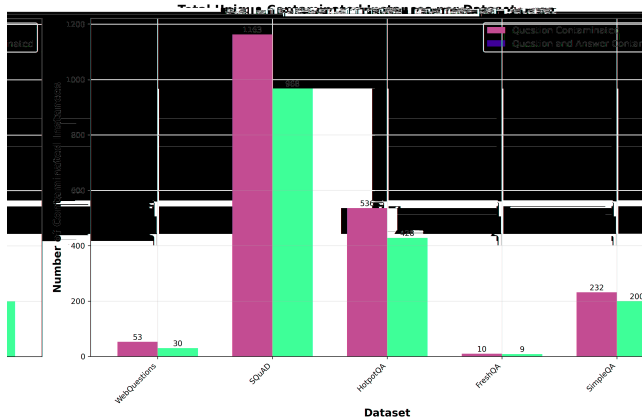


Figure 3: total_unique_retrieval_contaminated_instances_per_dataset.png

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